Lab 2 – Version 1

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October 26, 2026

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional requirements and overall characteristics of Gold Star. This document describes the product's intended capabilities, users, constraints and operating environment. It establishes a shared understanding of the software to be developed by the project team.

1.2 Scope

Gold Star is a web-based habit-building and time management platform designed for individuals with ADHD or ADHD-like traits who struggle to build habits and complete daily tasks. Gold Star uses Adaptive Scheduling to accommodate variability in the user's energy and productivity. The system encourages progress by rewarding completed tasks without penalizing for unfinished tasks. Gold Star does not diagnose or treat medical conditions and does not replace medical.

1.3 Definitions, Acronyms, and Abbreviations

This section contains the project team's glossary of terms, acronyms and abbreviations.

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions

allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

1.4 References

This section contains the project team's list of references.

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1.5 Overview

This document is organized into two main sections. Section 1 introduces the purpose and scope of the SRS and provides the project's glossary and references. Section 2 describes the product's perspective, primary functions, intended users, constraints, and assumptions and dependencies.

2 Gold Star Product Description

2.1 Product Perspective

Gold Star is a web-based habit-building platform designed for individuals struggling to maintain routines because of ADHD or ADHD-like traits. The application allows users to manage tasks and view their schedules while supporting changes to their daily plans.

Gold Star brings together calendar-based task organization, task prioritization, adaptive scheduling, habit stacking, and mood check-ins. These features are intended to help users organize responsibilities, establish routines, and respond to changes in their available time or productivity.

2.2 Main Product Functions

Gold Star provides task management and habit-building capabilities intended to help users plan daily responsibilities and maintain routines. Its main functions include:

- **Account Creation and Login:** Allow users to create an account and log-in to the application.

- **Calendar View:** Displays a monthly calendar, identifies days with scheduled tasks and displays scheduled tasks for selected day.
- **Daily Check-In:** Allows users to set and edit mood and productivity level for the day.
- **Adaptive Scheduling:** Allows users to reschedule unfinished tasks to another date and adjusts proposed tasks based on productivity level.
- **Task Prioritization:** Allows users to set task priority level as High, Medium or Low to help identify task importance level.
- **Habit Stacking:** Allows users to associate existing tasks so that one task can follow another as part of a routine.
- **Task Management:** Allows users to create, edit, remove and schedule tasks. A task includes a name, description, priority level, start time, end time and non-negotiable status.

2.3 User Characteristics

Scheduler

The Scheduler is the primary user of Gold Star. Schedulers use the application to create and manage tasks, organize their calendars, assign task priorities, adjust schedules, associate tasks into routines, and record their moods. The system is intended for users with varying levels of technical experience and provides a web-based graphical interface for interacting with its features.

Development Team

The development and evaluation team consists of the people responsible for implementing, testing, and evaluating Gold Star. Team members use this SRS to understand the expected behavior of the system and to verify that its implementation meets the documented requirements.

2.4 Constraints

N/A.

2.5 Assumptions and Dependencies

N/A.

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